

HENRY WILLIAMS

AI / AUTOMATION DEVELOPER • PYTHON & TYPESCRIPT

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PROFILE

Year 13 student and hands-on software builder focused on reliable AI agents, workflow automation and internal tools. I build end-to-end systems in Python and TypeScript, with particular emphasis on validation, auditability, idempotency, human-review gates, testing and explicit failure handling. Looking for bounded paid technical work where I can own a concrete component and prove value through delivery.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL • **Backend:** FastAPI, Fastify, Express, REST APIs, SQLite, PostgreSQL, OAuth • **AI & systems:** MCP/tool integration, agent workflows, local models, structured validation, audit logging • **Engineering:** pytest/testing, idempotency, failure handling, Git/GitHub, reproducible demos

SELECTED ENGINEERING WORK

Axiom AP Agent | *Python • payment controls • staging integration • 25-test suite* | [GitHub](#)

- Built and published an accounts-payable/payment agent with deterministic parsing and validation, explicit policy authority, human-review paths, audit JSONL and duplicate/idempotency protection.
- Engineered a staging adapter with redaction, request-ID correlation and replay/mismatch handling, plus a five-scenario verification runner covering allowed, policy-blocked and idempotency cases.
- Ran the staging verification against Axiom; their review confirmed the expected control behaviour, including policy blocks and no duplicate provider dispatch on replay. Kept staging evidence separate from any production claim.

Phone Command Bridge | *TypeScript • Fastify • SQLite • MCP*

- Designed an acknowledgement-based backend for tool-triggered phone actions so success is reported only after device confirmation, with durable state, leases, expiry, idempotency, hashed device tokens and OAuth verification.

BRANCH Operations System | *Python • SQLite • deterministic operations tooling*

- Built a business-operations CLI for customer, visit and payment records with deterministic reporting, malformed-input quarantine, verified backups and manual gates around external actions; 21-test local suite passed.

Spotify Playback Integration | *TypeScript • Express • PostgreSQL • OAuth*

- Built a bounded ChatGPT-controlled playback service with encrypted tokens, hashed grants, refresh rotation and explicit failure handling; source prepared for deployment without claiming public-production usage.

EDUCATION & DEVELOPMENT

Richard Challoner School, London - A Levels: Mathematics, Physics and German; Year 13	2025-2027
Study Mind Computer Science Summer Camp - frontend/backend development, algorithms and cybersecurity	2026
Harvard CS50x: Introduction to Computer Science - independent study	In progress

ADDITIONAL

Languages: English, French and German • **Availability:** around 10-14 hours/week during term for part-time or scoped remote technical work • **Other:** Duke of Edinburgh Bronze Award; 14 weeks of weekly cafe volunteering